Section-29

PORCELAIN LONG ROD INSULATORS

1.0 SCOPE : Refer Clause-1 of Section-I, General

This specification covers design, manufacture, testing at manufacturer's works before dispatch and supply of **Porcelain** Long Rod Insulators (Normal/**Anti fog**) as per the requirements given in Schedule of Requirements.

2.0 STANDARDS:

2.1 The insulators shall comply in all respects with the Indian standard specification IS-731-1976 , IS-3188 with latest amendments or IEC publication 383-1993, IEC 433-1980, IEC-575 ,IEC-797 or any other latest authoritative standard.

2.2 Insulators conforming to any other internationally accepted standards, which ensure equal or higher quality than the standard will also be acceptable. Where the material is offered according to any such international standard, an English version of the specification shall be attached to the tender.

3.0 ATMOSPHERIC CONDITIONS: Refer Clause-2 of Section-I, General

3.1 **SYSTEM CONDITIONS.**

Electrical system Data:

Electrical system Data:

a)	System Voltage (KV rms)	:	132/220
b)	Max. Voltage (KV rms)	:	145/245
c)	Lighting Impulse withstand voltage (Dry & Wet) (KVP)	:	650/1050
d)	Power frequency withstand voltage (wet) (KVP)	:	270/460

The materials offered shall be suitable for operation in tropical climate in which they will be subject to the full rays of the sun and inclement weather and shall be able to withstand wide range of temperature variations. The humidity may be as high as 100% during rainy season and as low as 10% during dry season. The prevailing weather conditions are given above.

3.2 Reference atmospheric conditions for which insulator characteristics shall be expressed for the purpose of comparison are given below.

Ambient temperature	20 Deg C
Barometric pressure	1013 Millibars
Absolute humidity	11 gms. of water per cubic Metre corresponding to 63 percent relative humidity at 20 Deg.C

3.3 Tests for the purpose of this specification may be carried out under conditions naturally obtaining at the time of the test. The barometric pressure, air temperature and humidity shall be recorded for the purpose of corrections.

4.0 GENERAL REQUIREMENTS:

4.1 All materials used in the manufacture of the Porcelain Long Rod insulators shall be of the best quality.

4.1.1. The glaze shall cover all porcelain parts of the Porcelain Long Rod insulators except those areas, which serve as supports during firing which are left unglazed for the purpose of assembly.

4.1.2. Cement used in the construction of Porcelain Long Rod insulators shall not cause fracture by expansion or loosening by contraction and proper care shall be taken to locate the individual parts correctly during cementing. The cement shall not give rise to chemical reaction with metal fittings and its thickness shall be as uniform as possible.

4.2 **DESIGN AND TYPE**

4.2.1 Type A insulators shall be offered. Type A insulator is one in which the length of the shortest puncture path through solid insulating material is more than half the length of the shortest flashover path through air outside the insulator.

4.2.2 The technical particulars of the Porcelain Long Rod insulators required and the system requirements are indicated in Annexure-I & clause '3.1' respectively. The design of insulators shall be such that stresses due to expansion and contraction in any part of the insulator shall not lead to deterioration.

4.2.3 The Porcelain Long Rod insulators shall be of ball and socket type. The ball and socket of the Porcelain Long Rod insulator shall be galvanized and mechanically strong. The ball shall move free in the sockets, but shall be so designed that they do not give way while in service.

4.2.4 The sockets shall be made of malleable iron. These shall be free from cracks, shinks air holes, burrs and rough edges. The socket shall be circular with inner and outer surfaces concentric and of such design that they will not yield or distort under the stress to the porcelain shells.

4.2.5 **MATERIALS:**

The ball pin shall be a single piece made of drop forged steel and shall be free from laps, folds, burrs and rough edges. All bearing surfaces shall be smooth and uniform so as to distribute the loading stresses uniformly. The ball pins shall be of such a design that they will not yield or distort under loaded condition. The pins shall not be made by joining, welding, shrink fitting or any other pieces from more than one piece of material.

4.2.6 Security clips for use with ball and socket coupling shall be in accordance with IS 2486 (Part III & IV) and made of suitable material of stainless steel or copper alloy or Phosphor-bronze. The security clips shall provide positive locking of the coupling. The logs of the security clips shall be spread after installation to prevent withdrawal from the socket.

4.2.7 The design of Long Rod Insulators shall be such that stresses due to expansion and contraction in any part of insulator shall not lead to deterioration. The porcelain shall not engage directly with hard metal.

PORCELAIN:

The porcelain used in the manufacture shall be sound, free from defects, thoroughly vitrified and smoothly glazed. The porcelain shall not engage directly with the hardware.

4.4 **GLAZE:**

The finished porcelain shall be glazed in brown colour. The glaze shall cover all exposed porcelain parts of the insulators, shall have a good luster, smooth surface and a good performance even under extreme weather conditions of tropical climate. It shall not get cracked or chipped by ageing under normal service conditions. The glaze shall have coefficient of expansion as that of the porcelain body throughout the working temperature range.

4.5 DIMENSIONS AND TOLERANCES:

The disc insulators covered by this specification shall have the dimensions specified in Annexure-I subject to the tolerances indicated in relevant IS.

WORKMANSHIP:

4.6.1 The insulators covered by this specification shall be of latest design and conform to the latest scientific methods and shall be suitable for use on 132 KV and 220 KV transmission lines, Sound design, latest manufacturing processes and proper material quality control shall be ensured at various stages. Good finish and elimination or sharp edges and corners shall also be ensured. The Porcelain Long Rod insulator shall be free from all sorts of defects. All exposed surfaces shall be smooth and perfect.

4.6.2 The forged metal sockets and shank pins shall be free from cracks, seams, shrinks, air-holes and rough edges. Metal pins shall be free from laps, folds, seams, burrs and rough edges. All surfaces of metal parts shall be perfectly smooth with no projecting points of irregularities.

4.6.3 All the ferrous parts shall be hot dip galvanized in accordance with the latest edition of IS 2629 and shall satisfy the tests mentioned in latest edition of IS 2633-1972. The zinc coating shall be adherent, smooth, reasonably bright, continuous and free from such imperfections like flux, ash, rust stains, bulky white deposits and blisters. Bituminous paint coating shall be provided on the metal units after assembly shall be concentric and co-axial within the limits as permitted by the relevant Indian. The zinc used for galvanizing shall be of grade Zn.99.95 as per IS-209-1979.

4.6.4 **SIZE AND NUMBER OF PORCELAIN LONG ROD :**

The size and number of Porcelain Long Rod to be provided in the insulator strings is indicated in Annexure-I.

INTER CHANGEABILITY:

The insulators inclusive of the ball and socket fittings shall be of standard design suitable for use with hardware fittings of any make conforming to relevant Indian Standards.

5.0 TESTING:

5.1 Suitable number of individual Porcelain Long Rod and accessories and complete strings shall be subjected to the tests as per latest editions of IS 731-1976, IS 2486 (Part-II)-1971, IEC 383 or any other equivalent authoritative standard.

5.2 TEST VOLTAGE:

The test voltage of the insulator shall be as given in Annexure-I.

5.3 MECHANICAL LOADS:

The insulator strings shall be suitable for the minimum failing loads specified in Annexure-I. The load shall be applied axially to the insulator strings.

5.4 TESTS:

TYPE TESTS:

The type tests must have been conducted not earlier than five years form the date of bid opening. The Bidder shall furnish one set of the following type test reports as per latest editions of IS:731-1976 , IEC-383-1993 IEC-433 and IEC-575 along with offer.

The original copies of relevant IEC / IS codes with latest revisions/amendments for reference make it available to APTRANSCO representatives during acceptance inspection tests and shall \be provided as soft copies

The Type test reports shall not be older than Five years(as per CEA regulation) and shall be valid till validity of offer.

The sample selection will be done as per Clause 14.9 and all the type test as per Clause 14.5.3 and 14.5.4 of this specification, are to be repeated on the offered lot without any extra cost to APTransco on random sample selected by APTRANSCO engineer at CPRI/ERDA or any NABL approved lab (as decided by APTRANSCO), in the presence of APTRANSCO engineer.

a) Visual examination
b) Verification of dimensions
c) Visible discharge test
d) Impulse voltage withstand test
e) Power frequency voltage withstand test i) dry and ii) wet and flashover test
f) Temperature cycle test
g) Electro-Mechanical failing load test
h) Puncture test **N.A.**
i) Porosity test
j) Galvanizing test

5.4.3 ROUTINE TESTS:

The following shall be carried out as routine tests.

a) Visual examination
b) Mechanical routine test
c) Electrical routine test (Not applicable for Long rod as the same is coming under Type A Puncture Proof)

5.4.4. ACCEPTANCE TESTS:

The test samples after having withstood the routine tests shall be subjected to the following acceptance tests as per latest edition of IS 731 in the order indicated below.

a) Verification of dimensions
b) Temperature cycle test
c) Twenty four hours mechanical strength test (Not applicable for Long rod as the same is coming under Type A Puncture Proof. Only Mechanical test applicable.)
d) Electro-Mechanical failing load test (This test is not applicable for Long rod as the same is coming under Type A Puncture Proof. Only Mechanical test applicable.)
e) Puncture test (Not applicable for Long rod as the same is coming under Type A Puncture Proof)
f) Porosity test and
g) Galvanizing test.

5.0 QUALITY ASSURANCE PLAN:

The Bidder shall invariably furnish following information along with his offer, failing which the offer shall be liable for rejection. Information shall be separately given for individual type of material offered.

i) Statement giving list of important raw materials, names of sub-suppliers for the raw material, list of standards according to which the raw materials are tested, list of tests normally carried out on raw materials in the presence of Bidders representative and copies of test certificates.
ii) Information and copies of test certificates as in (i) above in respect of bought out items.
iii) List of manufacturing facilities available.

6.0 MARKING:

7.1 Each insulator shall be legibly and indelibly marked to show the following:

a) Name or trade mark of the manufacturer
b) Month and year of manufacture
c) Minimum failing load in Kg or KN
d) Country of manufacture.

7.2 Markings on porcelain shall be printed and shall be applied before filing.

7.0 **CREEPAGE DISTANCES:**

The tenderer shall specify in the tender the creepage distance of the insulators. The minimum creepage distance required for insulators, to be used with their axis approximately vertical are given in Annexure – I.

9.0 INSPECTION:

The purchaser's representative shall have access to the manufacturer's works for purpose of inspection of the manufacture of the materials covered by this specification. 15 days notice shall invariably be given to the APTRANSCO for arranging inspection of the materials. The APTRANSCO will not be responsible for any delay if notice of 15 days is not given.

10.0 PACKING:

All insulators shall be packed in strong seasoned wooden crates suitably for easy but rough handling acceptable for rail and sea transport. When packed in a crate, as a string wooden separators shall be fixed between the insulators to keep individual insulator in position without movement within the crate.

11.0 GUARANTEED TECHNICAL PARTICULARS:

The number of disc insulators used on suspension and tension strings and the Electro mechanical strength are given in Annexure-I. Full guaranteed particulars including dry and wet flashovers, puncture and impulse voltages, corona formation voltages, creepage distances, length, voltage distribution on strings etc., shall be furnished.

12.0 SCHEDULE OF REQUIREMENTS AND DESIRED DELIVERY AND PRICES

12.1 The schedule of requirements and desired deliveries are indicated in Schedule of Requirements.

12.2. The schedule of prices should be indicated in the same Form

12.3 The prices shall be FIRM.

12.4 The requirement of materials indicated are only provisional and final quantities will be confirmed in the Purchase order.

13.0 DRAWINGS:

Fully dimensioned drawings shall be enclosed in duplicate to the tender, otherwise tender is liable for rejection. Drawing for purpose of tendering is enclosed.

ANNEXURE – I (Normal)
REQUIRED RATING AND ELECTRICAL CHARACTERISTICS

Sl. No.	Characteristics	Type of Insulators			
		132 kV Suspension String.	220 kV Suspension String	132 kV Tension String.	220 kV Tension String
1.	Type description	Long Rod Insulators	Long Rod Insulators	Long Rod Insulators	Long Rod Insulators
2.	No. of Porcelain Long rod insulator in a string	1	2	1	2
3.	Colour & surface of porcelain portion	BROWN GLAZED			
4.	Ferrous parts	AS PER IS-2629 & 2633 with latest editions			
5.	Check clip provided In the socket.	R type clip with stainless steel.	R type clip with stainless steel.	R type clip with stainless steel .	'R' type clip with stainless steel.
6.	Ball pin	Drop Forged Steel			
7.	Socket fitting	Malleable Cast Iron/Spherodical Graphite Iron			
8.	Ball pin designation	16 mm B	20mm	16 mm B	20mm
9. a)	Diameter and spacing of insulators (in mm).				
	Diameter	155	150	170	185
	Spacing	1305	1885	1450	2030
b)	Tolerances				
	(i) Dia +/-	7.5	7.5	8	8.5
	(ii) spacing +/-	38	53	42	56
10.	Minimum failing Load of the insulator (KN)	70	70	120	120
11.	Minimum total creepage distance in mm.	2655	3835	3200	4480
12.	Minimum protected creepage distance in mm	1065	1530	1280	1800
13.	Combined Mechanical and Electrical strength in KGs.	-	-	-	-
14.	One minute power Frequency withstand voltage (kV) Dry	325	460	325	510
15.	One minute power Frequency withstand voltage (kV) Wet	275	400	275	460
16.	Flash over Power Frequency voltage (kV)				
	Dry	350	490	350	560
	Wet	300	440	300	500
17.	Visible discharge voltage (kV)	105	154	105	154
18.	Lightening Impulse withstand voltage KVp	650	900	650	1050
19.	Impulse flash over voltage (kV)				
	Positive	700	980	700	1100
	Negative	720	1000	720	1120

20.	Standards according to which the porcelain insulator shall be manufactured and tested.	IS-731-1971 with latest amendments. IEC: 383, 433

ANNEXURE – I (Anti fog)

REQUIRED RATING AND ELECTRICAL CHARACTERISTICS

Sl. No.	Characteristics	Type of Insulators			
		132 kV Suspension String.	220 kV Suspension String	132 kV Tension String.	220 kV Tension String
1	Type description	Porcelain Long Rod Insulators	Porcelain Long Rod Insulators	Porcelain Long Rod Insulators	Porcelain Long Rod Insulators
2.	No. of Porcelain Long rod insulator in a string	1	2	1	2
3	Colour & surface of porcelain portion	BROWN GLAZED			
4	Ferrous parts	AS PER IS-2629 & 2633 with latest editions			
5	Check clip provided In the socket.	R type clip with stainless steel.	R type clip with stainless steel.	R type clip with stainless steel.	R' type clip with stainless steel.
6	Ball pin	Drop Forged Steel			
7	Socket fitting	Malleable Cast Iron/Spherodical Graphite Iron			
8	Ball pin designation	16 mm	20mm	16 mm	20mm
9 a)	Diameter and spacing of insulators (in mm).				
	Diameter	200	220	205	210
	Spacing	1305	1885	1450	2030
b)	Tolerances				
	(i) Dia +/-	9.5	10	9.5	10
	(ii) spacing +/-	38	53	42	56
10	Minimum failing Load of the insulator (KN)	90	90	120	120
11	Minimum total creepage distance in mm.	3915	5655	4350	6090
12	Minimum protected creepage distance in mm	1570	2265	1740	2440
13	Combined Mechanical and Electrical strength in KGs.	-	-	-	-
14	One minute power Frequency withstand voltage (kV) Dry	325	460	325	510
15	One minute power Frequency withstand voltage (kV) Wet	275	400	275	460
16	Flashover Power Frequency voltage (kV)				
	Dry	350	490	350	560
	Wet	300	440	300	500
17	Visible discharge voltage (kV)	105	154	105	154
18	Lightening Impulse withstand voltage KVp	650	900	650	1050
19	Impulse flash over voltage kV)				
	Positive	700	980	700	1100
	Negative	720	1000	720	1120

20	Standards according to which the porcelain insulator shall be manufactured and tested.	IS-731-1971 with latest amendments. IEC: 383, 433

Annexure - II

GUARANTEED PERFORMANCE & OTHER PARTICULARSFOR PORCELAIN LONG ROD INSULATORS BALL AND SOCKET TYPE FOR 220 KV AND 132 KV LINES

(While furnishing these details, the values in Annexure-I also are to be furnished.)

Sl.No.	Description	Electro Mechanical Strength			
		132 kV Suspension String	220 kV Suspension String	132 kV Tension String	220 kV Tension String
(1)	(2)	(3)	(4)	(5)	(6)
1.	a) Type of insulator (Ball and socket)				
	b) Size and designation ball and socket with standard to which it will conform.				
2.	Maker's name, address and country.				
3.	Dimensions:				
a)	Porcelain Long rod diameter in mm				
b)	Unit spacing in mm.				
c)	Creepage distance in mm.				
	i) Total				
	ii) Protected				
4.	Mechanical Values:				
a)	Combined Mechanical and Electrical strength in KGs.		NA		
b)	Mechanical breaking strength in KGs.				
c)	Routine Mechanical test load (KGs.)				
5.	Electrical Values:				
	a) Power frequency Flashover Voltage DRY (KV)				
	b) Power frequency Flashover Voltage WET (KV)				
	c) Impulse flashover voltage 1.2 x 40 mircro second negative wave (KV)				
	d) Impulse flashover voltage 1.2x40 mircro Second negativewave (KV)				
	e) Power frequency withstand voltage : DRY (KV)				
	f) Power frequency withstand voltage : WET (KV)				
	g) Impulse withstand test voltage 1.2x40 micro second positive and negative wave(KV)				
	h) Power frequency puncture Voltage (KV)		NA		
6.	Net weight of insulator string in KGs.				
7.	Strandard specification to				

which insulators will conform.

8. Other particulars if any

SIGNATURE OF THE TENDERER